

Hallucination Detection Report

AI Output Verification · Knowledge Graph Analysis

Generated: 2026-04-16 15:30
Triples: 8 · Verdicts: 5 · Inconsistencies: 0 ·
Violations: 3

EXECUTIVE SUMMARY

2 Correct	1 Predicate Mismatch	2 Fabricated Fact
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EXTRACTED TRIPLES

All verifiable triples extracted from the LLM response before canonicalization or checking.

Subject	Predicate	Object	Verifiable	Verification Type	Predicate Definition
575	HAS_INTEREST_RATE	12.21%	Yes	ENTITY_ATTRIBUTE	Indicates the percent-based interest rate associated with the subject entity.
580	HAS_INTEREST_RATE	11.91%	Yes	ENTITY_ATTRIBUTE	Indicates the percent-based interest rate associated with the subject entity.
579	HAS_INTEREST_RATE	12.03%	Yes	ENTITY_ATTRIBUTE	Indicates the percent-based interest rate associated with the subject entity.
579	HAS_RISK_TIER	deep subprime	Yes	ENTITY_ATTRIBUTE	Specifies the risk level tier classified for the subject entity.
890	HAS_INTEREST_RATE	Below 4.36%	Yes	ENTITY_ATTRIBUTE	Indicates that the interest rate for the subject entity is lower than 4.36%.
890	HAS_RISK_TIER	super prime	Yes	ENTITY_ATTRIBUTE	Specifies the risk level tier classified for the subject entity.
890	HAS_DOWN_PAYMENT	Below 2.0%	Yes	ENTITY_ATTRIBUTE	Indicates that the down payment for the subject entity is lower than 2.0%.
335	HAS_LOAN_TERM	12 months	Yes	ENTITY_ATTRIBUTE	Indicates the loan term duration for the subject entity.

CANONICALIZED TRIPLES

Predicates normalized against the database schema. Shows original predicate, canonical match, and match confidence.

Subject	Original Predicate	Object	Canonical Predicate	Confidence
575	HAS_INTEREST_RATE	12.21%	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
580	HAS_INTEREST_RATE	11.91%	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
579	HAS_INTEREST_RATE	12.03%	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
579	HAS_RISK_TIER	deep subprime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
890	HAS_INTEREST_RATE	Below 4.36%	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
890	HAS_RISK_TIER	super prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
890	HAS_DOWN_PAYMENT	Below 2.0%	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	HIGH

Subject	Original Predicate	Object	Canonical Predicate	Confidence
335	HAS_LOAN_TERM	12 months	LOAN_TERM_GRANTED_AS_MAXIMUM_REPAYMENT_WINDOW_OFFERED	HIGH

LOGICAL INCONSISTENCIES

Contradictions detected between triples within the LLM response, independent of the database.

No logical inconsistencies detected.

SYMBOLIC RULE VIOLATIONS

Violations caught by explicit domain constraints before database lookup. Violated triples were removed from the evidence retrieval step.

c_interest_rate_range • • MEDIUM

890 → INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE → Below 4.36%

Credit score 890 is above the allowed range (850), which violates the constraint regardless of interest rate.

c_tier_300_579 • • MEDIUM

890 → RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL → super prime

Credit score 890 is above the allowed range (850), which violates the constraint regardless of risk tier.

c_down_payment_range • • MEDIUM

890 → DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION → Below 2.0%

Credit score 890 is above 850, which violates the constraint regardless of down payment.

DETECTED HALLUCINATIONS

Subject	Predicate	Object	Verdict	Match	Violated	Conf.	Explanation
575	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	12.21%	Correct	STRICT	—	1.00	The subject, predicate, and numeric value (ignoring % symbol) match exactly with the DB candidate. Evidence: subject: 575, predicate: INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE, object: 12.21
580	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	11.91%	Correct	STRICT	—	1.00	The subject, predicate, and numeric value (ignoring % symbol) match exactly with the DB candidate. Evidence: subject: 580, predicate: INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE, object: 11.91
579	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	12.03%	Fabricated Fact	NONE	SUBJECT	1.00	No DB candidate has subject 579 with this predicate; the assertion is fabricated.

Subject	Predicate	Object	Verdict	Match	Violated	Conf.	Explanation
579	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	deep subprime	Fabricated Fact	NON E	SUBJECT	0.60	No DB candidate links subject 579 to 'deep subprime'; the subject is not present in any triple for this predicate. Evidence: subject: 375, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 380, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 385, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 390, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 395, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 400, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 405, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 410, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 415, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 420, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 425, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 430, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 435, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 440, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 445, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime
335	LOAN_TERM_GRANTED_AS_MAXIMUM_REPAYMENT_WINDOW_OFFERED	12 months	Predicate Mismatch	NON E	PREDICATE	0.60	Subject 335 exists, but the predicate 'LOAN_TERM_GRANTED_AS_MAXIMUM_REPAYMENT_WINDOW_OFFERED' is not present for this subject in DB. Evidence: subject: 335, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 335, predicate: INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE, object: 20.88

TRIPLE AUDIT TRAIL — ORIGINAL → CANONICAL → VERDICT

Full pipeline trace for every triple. UNVERIFIABLE* means the triple had no verdict — e.g. dropped due to output truncation.

Orig. Subject	Orig. Predicate	Orig. Object	Confidence	Canon. Predicate	Canon. Object	Verdict
575	HAS_INTEREST_RATE	12.21%	HIGH	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	12.21%	CORRECT
580	HAS_INTEREST_RATE	11.91%	HIGH	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	11.91%	CORRECT
579	HAS_INTEREST_RATE	12.03%	HIGH	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	12.03%	FABRICATED_FACT
579	HAS_RISK_TIER	deep subprime	HIGH	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	deep subprime	FABRICATED_FACT
335	HAS_LOAN_TERM	12 months	HIGH	LOAN_TERM_GRANTED_AS_MAXIMUM_REPAYMENT_WINDOW_OFFERED	12 months	PREDICATE_MISMATCH

VERDICT LEGEND

Verdict	Description
Correct	All three components (subject, predicate, object) verified against DB.
Rule Violation	One or more components contradict DB evidence.
Multi-Rule Violation	Violation spans multiple DB triples simultaneously.
Unverifiable	No usable DB evidence found for any component.
Predicate Mismatch	Subject real, but stated predicate is not a valid relation for it.
Fabricated Fact	Triple asserts a relationship that does not exist anywhere in DB.